

Initiating Search

November 6, 2025, 4:26 PM

• Search:

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Acetals - Preferred Concept

Search Tasks

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Returned Reference Results from CAS Lexicon (9,638)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (2,890)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%	
Document Type:	Journal	
Language:	English	

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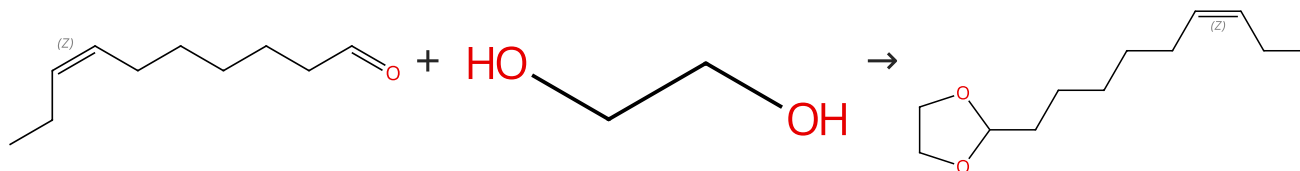
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

 Suppliers (166)

Double bond geometry shown

 Suppliers (25)

 Supplier (1)

31-301-CAS-1879451

Steps: 1 Yield: 100%

1,3-Dioxolanes from carbonyl compounds over zeolite HSZ-360 as a reusable, heterogeneous catalyst

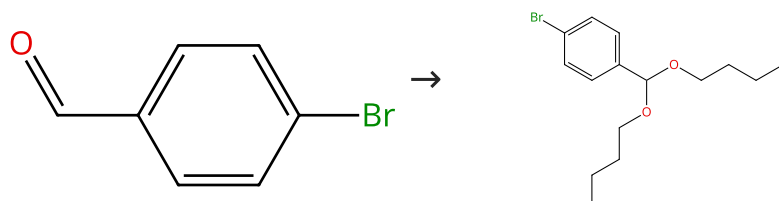
By: Ballini, Roberto; et al

Tetrahedron Letters (1998), 39(12), 1615-1618.

1.1 Solvents: Toluene

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (91)

 Suppliers (3)

31-301-CAS-3144520

Steps: 1 Yield: 100%

Polymer-supported Lewis acid catalysts. I. Polystyrene-gallium trichloride complex

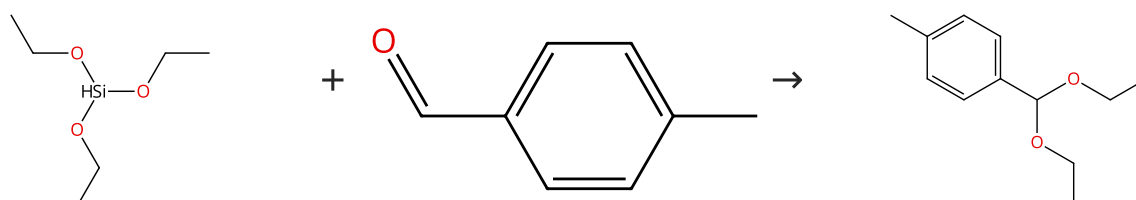
By: Ruicheng, Ran; et al

Journal of Macromolecular Science, Chemistry (1987), A24(6), 669-79.

1.1 Reagents: 1-Butanol
Catalysts: Gallium chloride (polymer bound)
Solvents: Benzene

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (23)

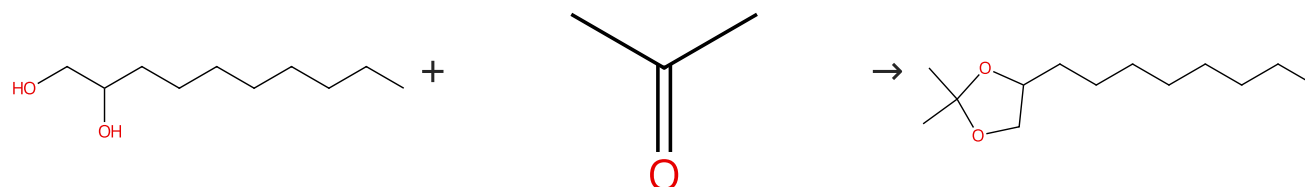
 Suppliers (104)

 Suppliers (24)

31-301-CAS-17073938	Steps: 1 Yield: 100%	Antimony(V) catalyzed acetalisation of aldehydes: an efficient, solvent-free, and recyclable process
1.1 Catalysts: 1959623-98-3; rt		By: Ugarte, Renzo Arias; et al
1.2 rt; 1.5 h, rt		Green Chemistry (2017), 19(8), 1990-1998.
Experimental Protocols		

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (60)

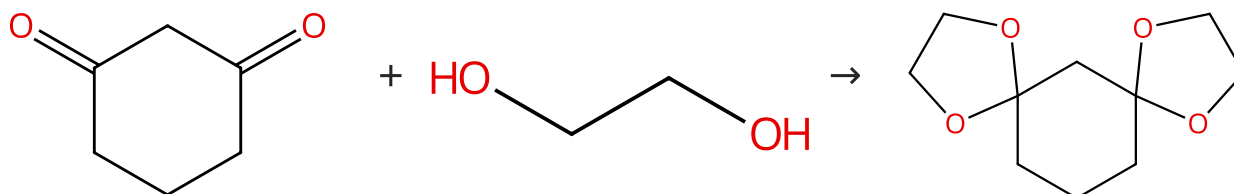
Suppliers (296)

Supplier (1)

31-301-CAS-18007949	Steps: 1 Yield: 100%	Utilization of hexabromoacetone for protection of alcohols and aldehydes and deprotection of acetals, ketals, and oximes under UV irradiation
1.1 Catalysts: 1,1,1,3,3,3-Hexabromo-2-propanone; 10 min, rt		By: Chaiseeda, Kittichai; et al
Experimental Protocols		Research on Chemical Intermediates (2018), 44(2), 1305-1323.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (97)

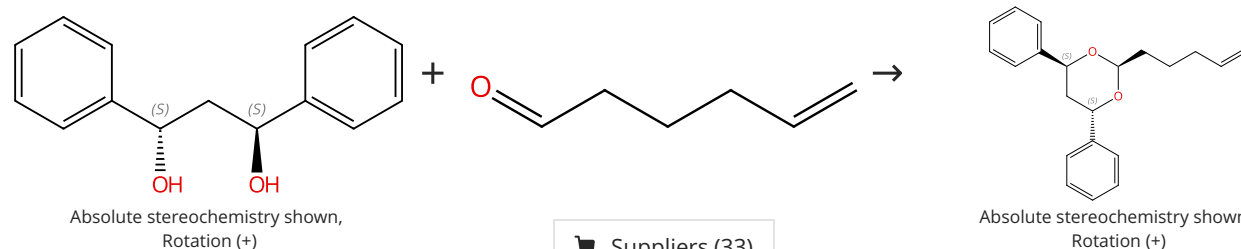
Suppliers (166)

Suppliers (4)

31-301-CAS-4265795	Steps: 1 Yield: 100%	1,3-Dioxolanes from carbonyl compounds over zeolite HSZ-360 as a reusable, heterogeneous catalyst
1.1 Solvents: Toluene		By: Ballini, Roberto; et al
		Tetrahedron Letters (1998), 39(12), 1615-1618.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (7)

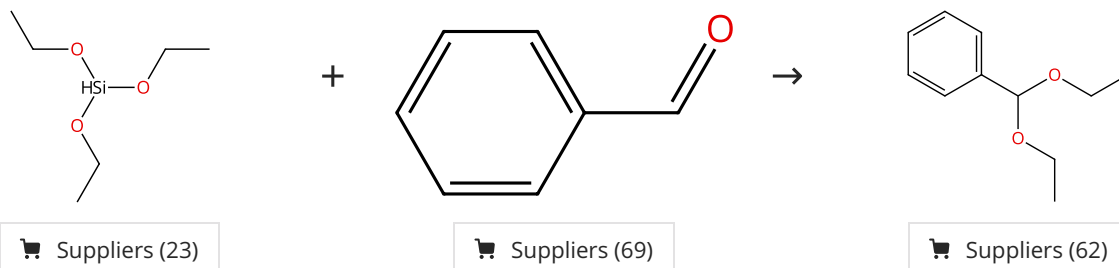
Suppliers (33)

Absolute stereochemistry shown,
Rotation (+)

31-301-CAS-5572124	Steps: 1 Yield: 100%	Asymmetric Induction via an Intramolecular Haloetherification Reaction of Chiral Ene Acetals: A Novel Approach to Optically Active 1,4- and 1,5-Diols By: Fujioka, Hiromichi; et al Journal of Organic Chemistry (1996), 61(21), 7309-7315.
1.1 Reagents: <i>p</i> -Toluenesulfonic acid Solvents: Benzene		

Scheme 7 (1 Reaction)

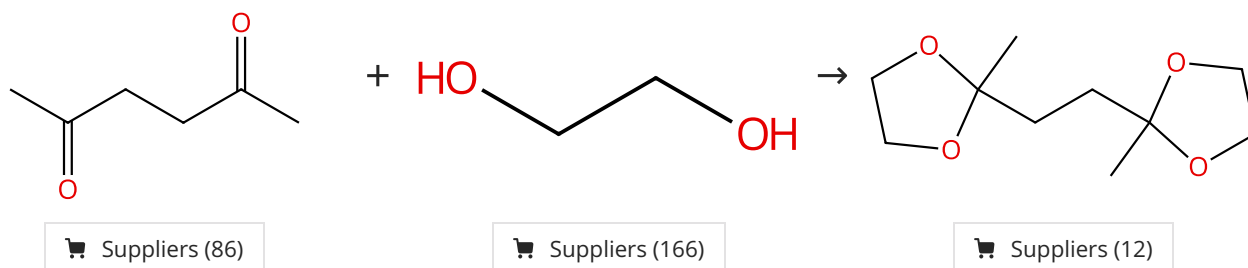
Steps: 1 Yield: 100%



31-301-CAS-17073916	Steps: 1 Yield: 100%	Antimony(V) catalyzed acetalisation of aldehydes: an efficient, solvent-free, and recyclable process By: Ugarte, Renzo Arias; et al Green Chemistry (2017), 19(8), 1990-1998.
1.1 Catalysts: 1959623-98-3; rt		
1.2 rt; 0.16 h, rt		
Experimental Protocols		

Scheme 8 (1 Reaction)

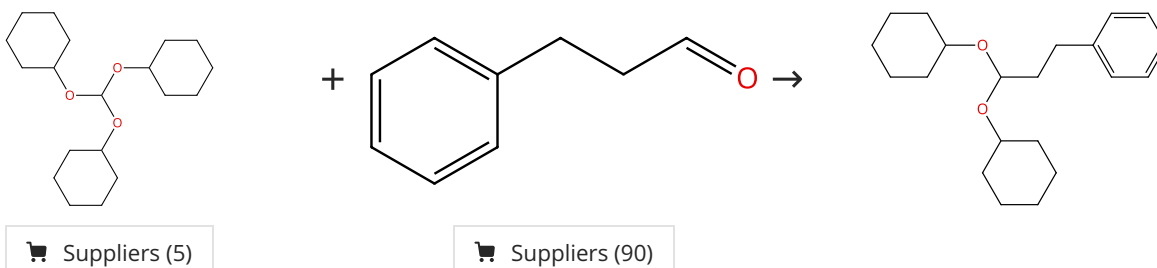
Steps: 1 Yield: 100%



31-301-CAS-12810656	Steps: 1 Yield: 100%	1,3-Dioxolanes from carbonyl compounds over zeolite HSZ-360 as a reusable, heterogeneous catalyst By: Ballini, Roberto; et al Tetrahedron Letters (1998), 39(12), 1615-1618.
1.1 Solvents: Toluene		

Scheme 9 (1 Reaction)

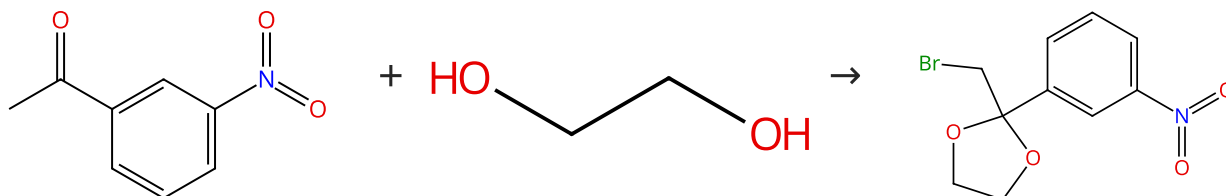
Steps: 1 Yield: 100%



31-478-CAS-12005391 Steps: 1 Yield: 100% 1.1 Catalysts: Methanesulfonic acid, 1,1,1-trifluoro-, cerium(3+) salt (3:1) Solvents: Toluene, Cyclohexanol; 20 min, rt 1.2 Reagents: Triethylamine Experimental Protocols	A simple and efficient procedure for the synthesis of ketone di-sec-alkyl acetals By: Ono, Fumiaki; et al Synlett (2009), (3), 487-489.
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Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

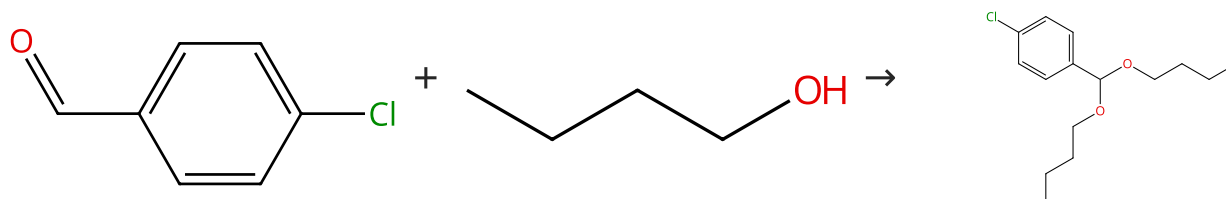
Suppliers (166)

Suppliers (4)

31-301-CAS-16461483 Steps: 1 Yield: 100% 1.1 Reagents: <i>N</i>-Bromosuccinimide Solvents: Ethylene glycol; 1 h, rt; 24 h, 45 °C Experimental Protocols	One-step for the preparation of α-haloacetal of ketones with <i>N</i>-bromosuccinimide/<i>N</i>-chlorosuccinimide (NBS/NCS) and ethylene glycol By: Zheng, Zubiao; et al Tetrahedron (2016), 72(48), 7738-7743.
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Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (99)

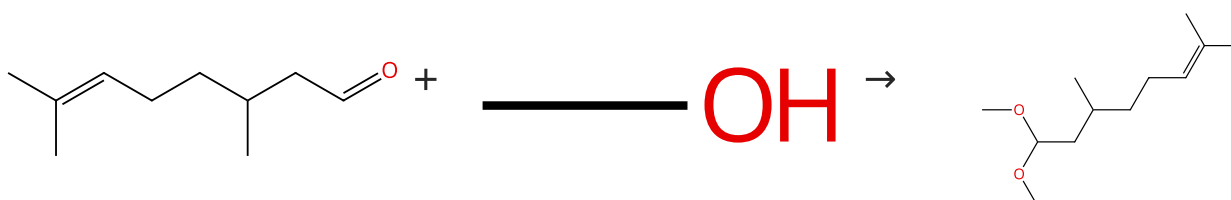
Suppliers (201)

Suppliers (2)

31-301-CAS-11556431 Steps: 1 Yield: 100% 1.1 Catalysts: <i>p</i>-Toluenesulfonic acid monohydrate Solvents: Toluene, Water; 4 h, rt 1.2 Reagents: Triethylamine; rt	Can Molecular Sieves be Used as Water Scavengers in Microwave Chemistry? By: Baghbanzadeh, Mostafa; et al Australian Journal of Chemistry (2009), 62(3), 244-249.
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Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (89)

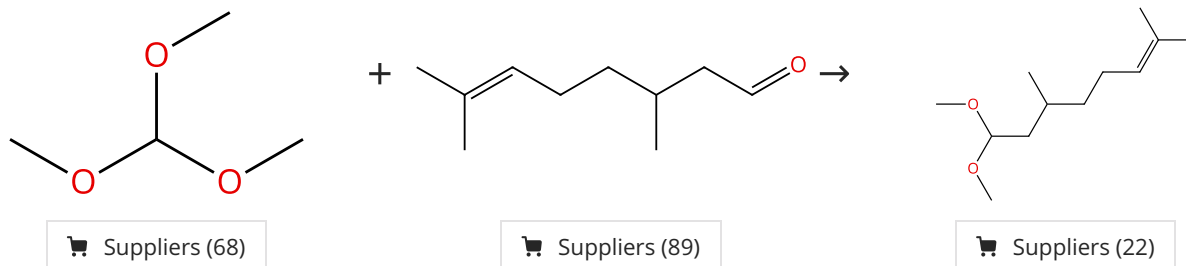
Suppliers (444)

Suppliers (22)

31-301-CAS-5205328	Steps: 1 Yield: 100%	An efficient and versatile procedure for the synthesis of acetals from aldehydes and ketones catalyzed by lithium tetrafluoroborate
1.1 Reagents: Trimethyl orthoformate Catalysts: Lithium tetrafluoroborate Solvents: Methanol; 2 h, reflux		By: Hamada, Nao; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water		Synlett (2004), (6), 1074-1076.
Experimental Protocols		

Scheme 13 (1 Reaction)

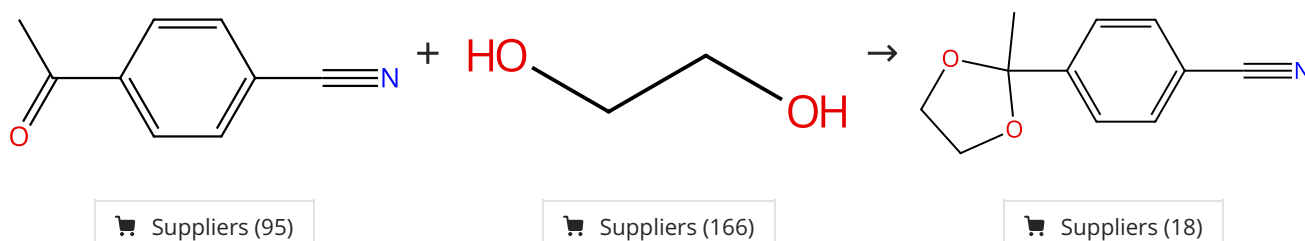
Steps: 1 Yield: 100%



31-614-CAS-43207917	Steps: 1 Yield: 100%	Diastereoselective Substitution Reactions of Acyclic Acetals Controlled by Remote Participation of an Acyloxy Group
1.1 Reagents: <i>p</i> -Toluenesulfonic acid Solvents: Methanol; 22 °C		By: Luu, Khoi B.; et al
Experimental Protocols		Organic Letters (2024), 26(49), 10470-10474.

Scheme 14 (1 Reaction)

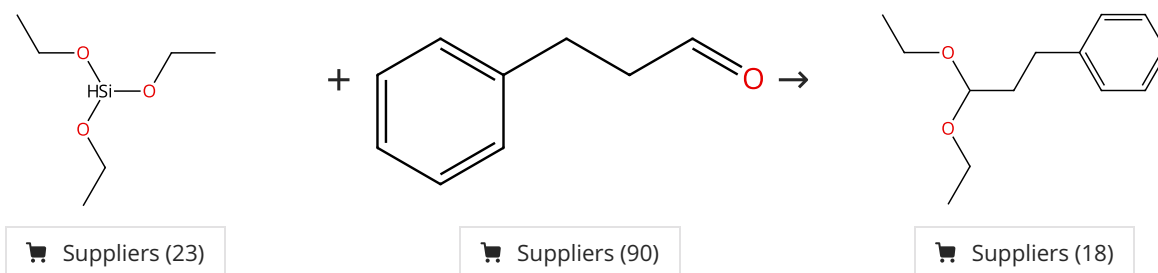
Steps: 1 Yield: 100%



31-301-CAS-11009444	Steps: 1 Yield: 100%	1,3-Dioxolanes from carbonyl compounds over zeolite HSZ-360 as a reusable, heterogeneous catalyst
1.1 Solvents: Toluene		By: Ballini, Roberto; et al
		Tetrahedron Letters (1998), 39(12), 1615-1618.

Scheme 15 (1 Reaction)

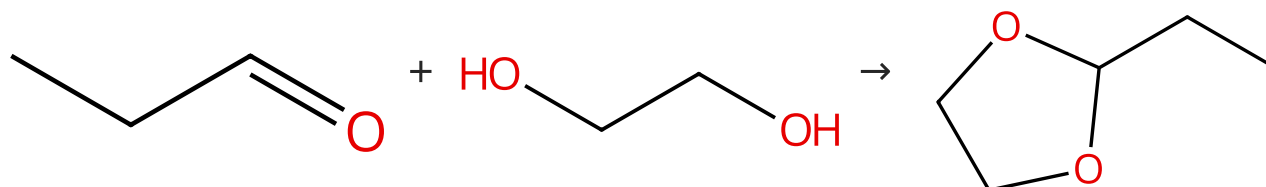
Steps: 1 Yield: 100%



31-301-CAS-17073913	Steps: 1 Yield: 100% Antimony(V) catalyzed acetalisation of aldehydes: an efficient, solvent-free, and recyclable process
1.1 Catalysts: 1959623-98-3; rt 1.2 rt; 0.16 h, rt	By: Ugarte, Renzo Arias; et al
Experimental Protocols	Green Chemistry (2017), 19(8), 1990-1998.

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

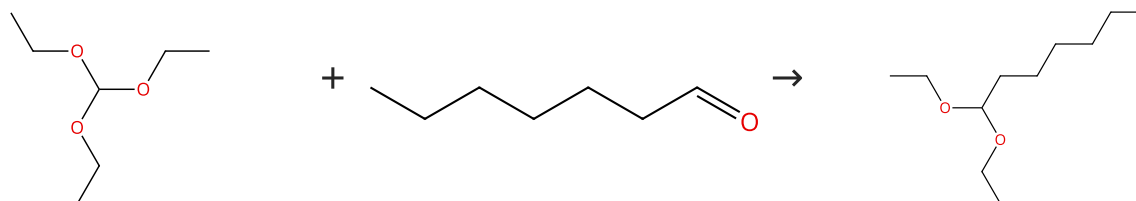
Suppliers (166)

Suppliers (28)

31-301-CAS-12445934	Steps: 1 Yield: 100% Highly efficient and versatile acetalization of glycol catalyzed by cupric p-toluenesulfonate
1.1 Catalysts: Benzenesulfonic acid, 4-methyl-, copper(2+) salt (2: 1); 90 min, reflux	By: Liu, Cai-Hua; et al
	Chinese Journal of Chemistry (2007), 25(1), 83-85.

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (71)

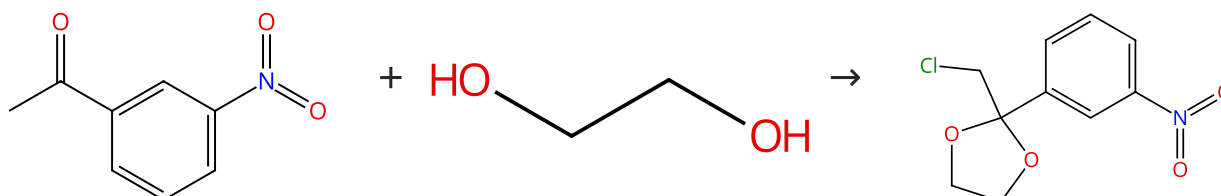
Suppliers (65)

Suppliers (20)

31-301-CAS-23962778	Steps: 1 Yield: 100% Practical acetalization and transacetalization of carbonyl compounds catalyzed by recyclable PVP-I
1.1 Catalysts: Poly(vinylpyrrolidone) hydrogen triiodide Solvents: Ethanol; 3 h, 30 °C	By: Wang, Di; et al
Experimental Protocols	Tetrahedron (2021), 92, 132250.

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

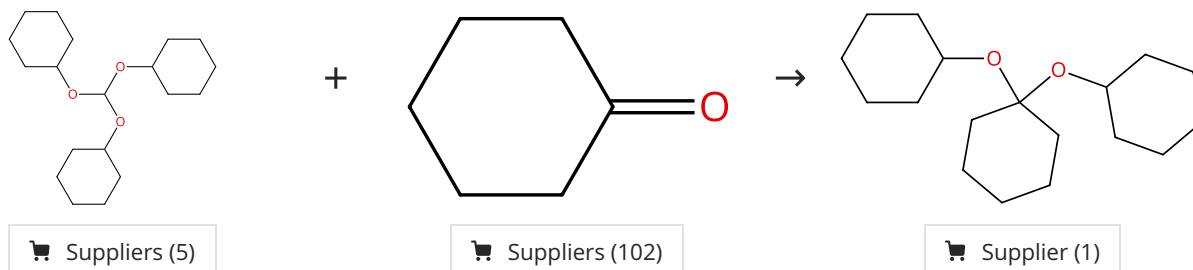
Suppliers (166)

Suppliers (2)

31-301-CAS-16461503	Steps: 1 Yield: 100%	One-step for the preparation of α-haloacetal of ketones with N-bromosuccinimide/N-chlorosuccinimide (NBS/NCS) and ethylene glycol By: Zheng, Zubiao; et al Tetrahedron (2016), 72(48), 7738-7743.
1.1 Reagents: Chlorosuccinimide Solvents: Ethylene glycol; 1 h, rt; 24 h, 45 °C		
Experimental Protocols		

Scheme 19 (1 Reaction)

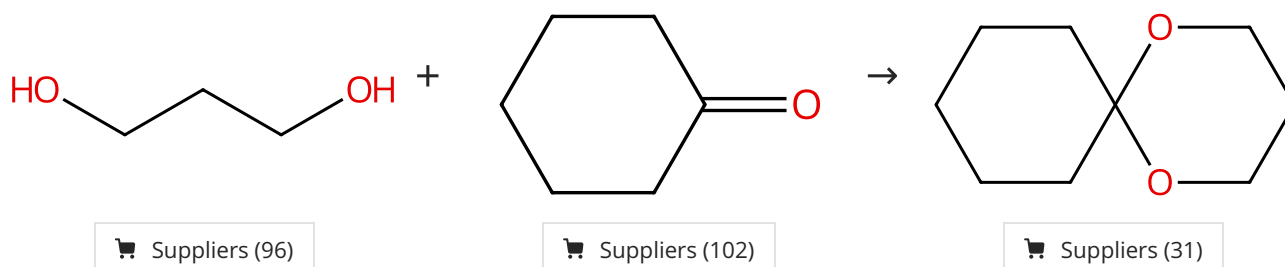
Steps: 1 Yield: 100%



31-301-CAS-7751715	Steps: 1 Yield: 100%	A simple and efficient procedure for the synthesis of ketone di-sec-alkyl acetals By: Ono, Fumiaki; et al Synlett (2009), (3), 487-489.
1.1 Catalysts: Methanesulfonic acid, 1,1,1-trifluoro-, cerium(3+) salt (3:1) Solvents: Toluene, Cyclohexanol; 10 min, rt 1.2 Reagents: Triethylamine		
Experimental Protocols		

Scheme 20 (1 Reaction)

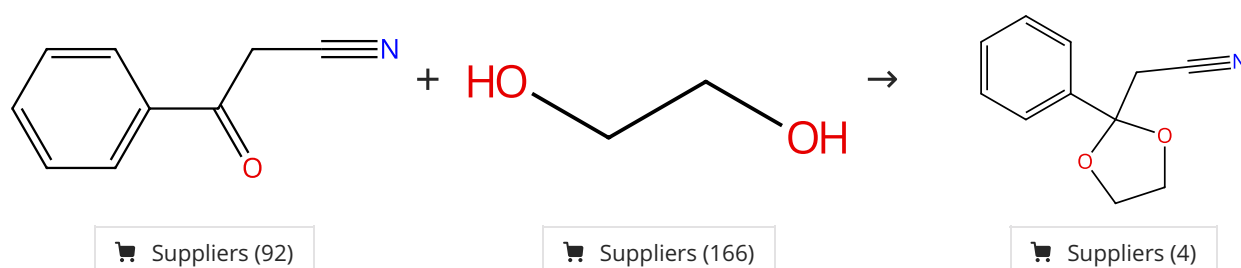
Steps: 1 Yield: 100%



31-301-CAS-1979005	Steps: 1 Yield: 100%	2,4,4,6-Tetrabromo-2,5-cyclohexadienone (TABCO) as a versatile, efficient, and chemoselective catalyst for the acetalization and transacetalization of carbonyl compounds, the preparation of acetonides from epoxides and acylals (1, 1-diacetates) from aldehydes By: Firouzabadi, Habib; et al Bulletin of the Chemical Society of Japan (2002), 75(10), 2195-2205.
1.1 Reagents: Triethyl orthoformate Catalysts: 2,4,4,6-Tetrabromo-2,5-cyclohexadien-1-one Solvents: Ethanol; 25 h, 50 °C		
1.2 Reagents: Sodium hydroxide Solvents: Water		

Scheme 21 (1 Reaction)

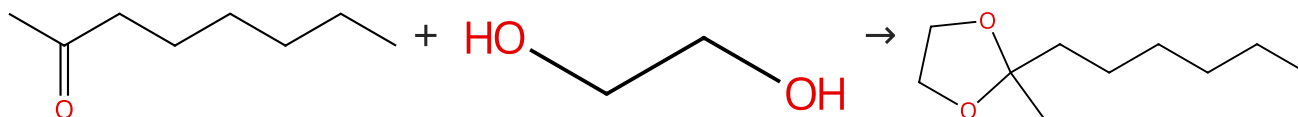
Steps: 1 Yield: 100%



31-614-CAS-43501703	Steps: 1 Yield: 100%	Mg(OTf)₂·SiO₂: a heterogeneous, recyclable catalyst for the synthesis of acetals from aldehydes or ketones
1.1 Reagents: <i>p</i> -Toluenesulfonic acid Catalysts: Magnesium triflate (supported on silica) Solvents: Toluene; rt → 115 °C; 6 - 8 h, 115 °C		By: More, Mahesh; et al
Experimental Protocols		Heterocyclic Letters (2024), 14(4), 787-795.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

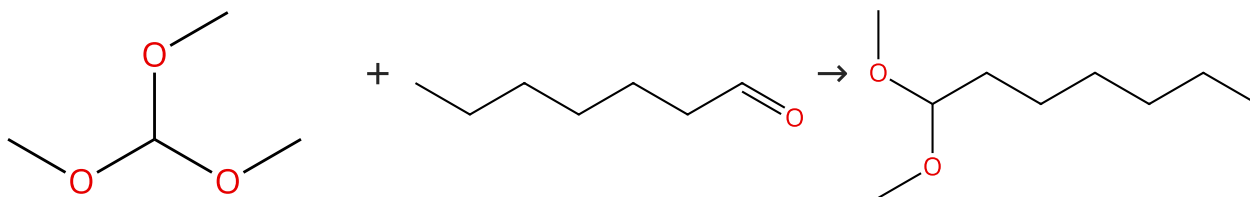
Suppliers (166)

Suppliers (7)

31-301-CAS-13973641	Steps: 1 Yield: 100%	Unusual rate acceleration in the presence of aldehydes of distannoxane-catalyzed acetalization of ketones and deactivated aldehydes: a new mode of carbonyl activation
1.1 Catalysts: Octabutyl-di-μ-hydroxydi-μ ₃ -oxobis(thiocyanato-κM) tetratin Solvents: Benzene		By: Otera, Junzo; et al
		Organometallics (1989), 8(8), 2063-5.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (68)

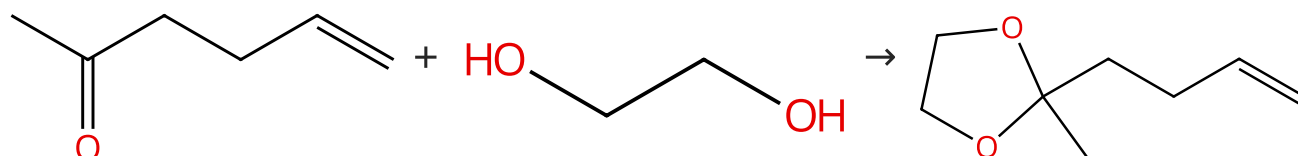
Suppliers (65)

Suppliers (35)

31-301-CAS-23963948	Steps: 1 Yield: 100%	Practical acetalization and transacetalization of carbonyl compounds catalyzed by recyclable PVP-I
1.1 Catalysts: Poly(vinylpyrrolidone) hydrogen triiodide Solvents: Methanol; 3 h, 30 °C		By: Wang, Di; et al
Experimental Protocols		Tetrahedron (2021), 92, 132250.

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (57)

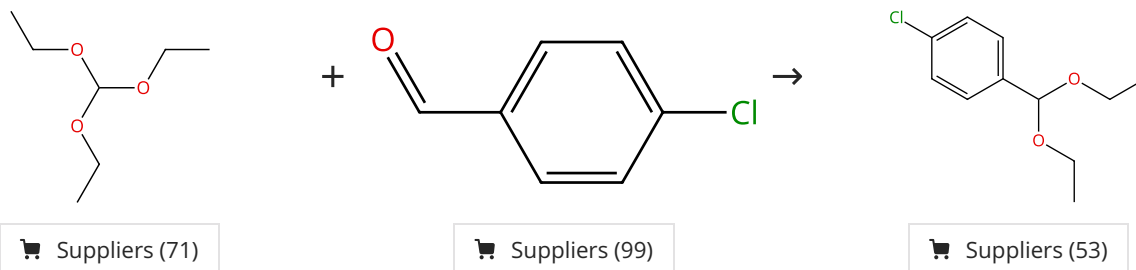
Suppliers (166)

Supplier (1)

31-301-CAS-13420797	Steps: 1 Yield: 100%	1,3-Dioxolanes from carbonyl compounds over zeolite HSZ-360 as a reusable, heterogeneous catalyst
1.1 Solvents: Toluene		By: Ballini, Roberto; et al Tetrahedron Letters (1998), 39(12), 1615-1618.

Scheme 25 (1 Reaction)

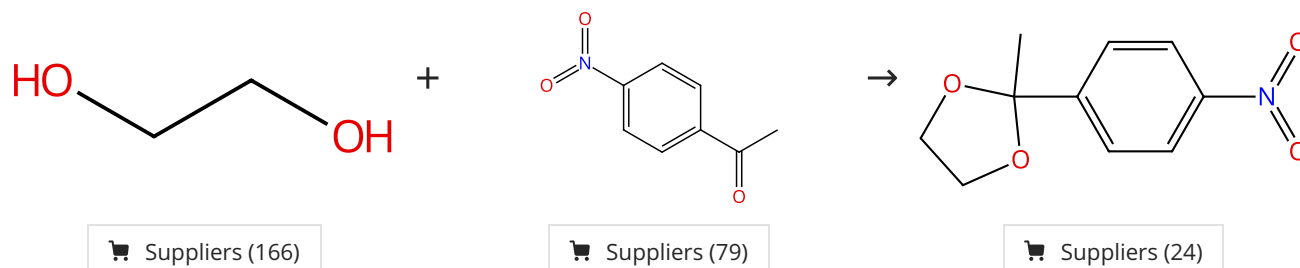
Steps: 1 Yield: 100%



31-301-CAS-23963920	Steps: 1 Yield: 100%	Practical acetalization and transacetalization of carbonyl compounds catalyzed by recyclable PVP-I
1.1 Catalysts: Poly(vinylpyrrolidone) hydrogen triiodide Solvents: Ethanol; 3 h, 30 °C		By: Wang, Di; et al Tetrahedron (2021), 92, 132250.
Experimental Protocols		

Scheme 26 (1 Reaction)

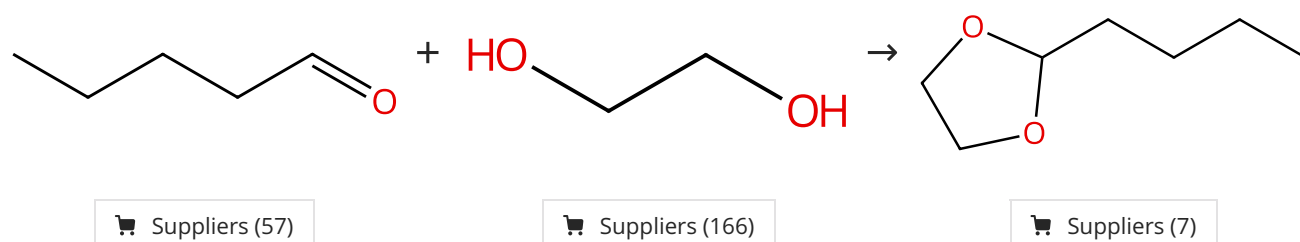
Steps: 1 Yield: 100%



31-301-CAS-8872523	Steps: 1 Yield: 100%	1,3-Dioxolanes from carbonyl compounds over zeolite HSZ-360 as a reusable, heterogeneous catalyst
1.1 Solvents: Toluene		By: Ballini, Roberto; et al Tetrahedron Letters (1998), 39(12), 1615-1618.

Scheme 27 (1 Reaction)

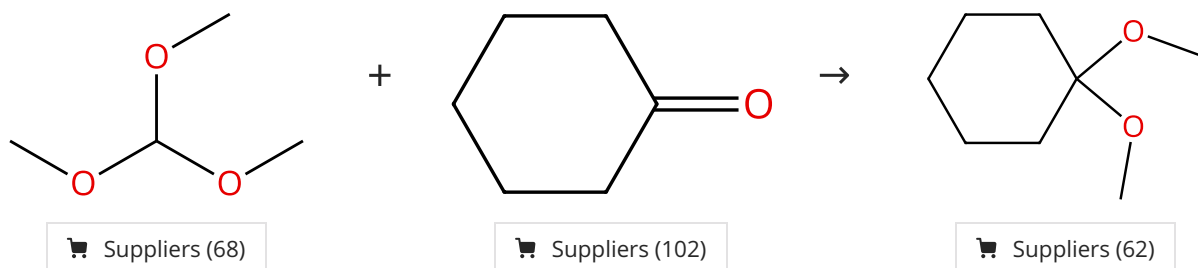
Steps: 1 Yield: 100%



31-301-CAS-3629092	Steps: 1 Yield: 100%	Highly efficient and versatile acetalization of glycol catalyzed by cupric p-toluenesulfonate
1.1 Catalysts: Benzenesulfonic acid, 4-methyl-, copper(2+) salt (2:1); 90 min, reflux		By: Liu, Cai-Hua; et al Chinese Journal of Chemistry (2007), 25(1), 83-85.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 100%



31-301-CAS-23964167

Steps: 1 Yield: 100%

Practical acetalization and transacetalization of carbonyl compounds catalyzed by recyclable PVP-I

1.1 Catalysts: Poly(vinylpyrrolidone) hydrogen triiodide
Solvents: Methanol; 6 h, 30 °C

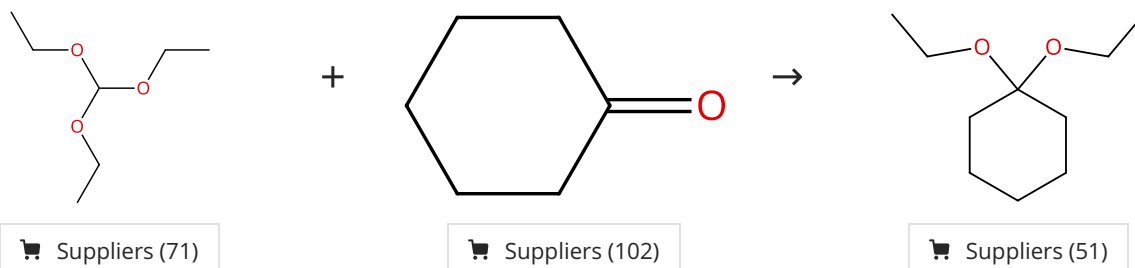
By: Wang, Di; et al

Experimental Protocols

Tetrahedron (2021), 92, 132250.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 100%



31-301-CAS-23963266

Steps: 1 Yield: 100%

Practical acetalization and transacetalization of carbonyl compounds catalyzed by recyclable PVP-I

1.1 Catalysts: Poly(vinylpyrrolidone) hydrogen triiodide
Solvents: Ethanol; 6 h, 30 °C

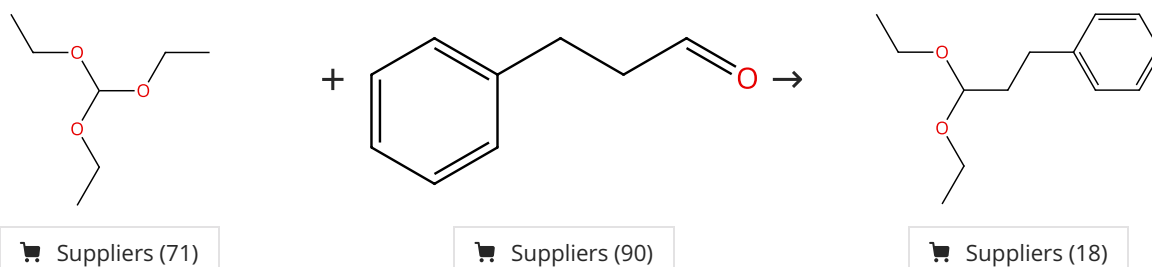
By: Wang, Di; et al

Experimental Protocols

Tetrahedron (2021), 92, 132250.

Scheme 30 (2 Reactions)

Steps: 1 Yield: 100%



31-301-CAS-23964149

Steps: 1 Yield: 100%

Practical acetalization and transacetalization of carbonyl compounds catalyzed by recyclable PVP-I

1.1 Catalysts: Poly(vinylpyrrolidone) hydrogen triiodide
Solvents: Ethanol; 3 h, 30 °C

By: Wang, Di; et al

Experimental Protocols

Tetrahedron (2021), 92, 132250.

31-301-CAS-21315153

Steps: 1 Yield: 100%

Antimony(V) catalyzed acetalisation of aldehydes: an efficient, solvent-free, and recyclable process

1.1 Catalysts: 1959623-98-3; rt
1.2 rt; 12 h, rt

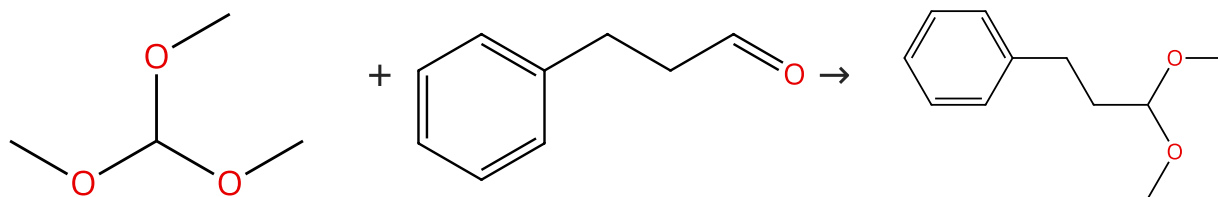
By: Ugarte, Renzo Arias; et al

Experimental Protocols

Green Chemistry (2017), 19(8), 1990-1998.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (68)

Suppliers (90)

Suppliers (13)

31-301-CAS-23963740

Steps: 1 Yield: 100%

Practical acetalization and transacetalization of carbonyl compounds catalyzed by recyclable PVP-I

1.1 Catalysts: Poly(vinylpyrrolidone) hydrogen triiodide
Solvents: Methanol; 3 h, 30 °C

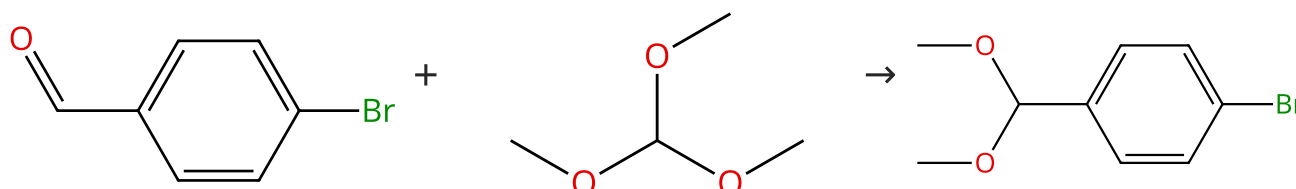
By: Wang, Di; et al

Experimental Protocols

Tetrahedron (2021), 92, 132250.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (91)

Suppliers (68)

Suppliers (61)

31-301-CAS-23962650

Steps: 1 Yield: 100%

Practical acetalization and transacetalization of carbonyl compounds catalyzed by recyclable PVP-I

1.1 Catalysts: Poly(vinylpyrrolidone) hydrogen triiodide
Solvents: Methanol; 3 h, 30 °C

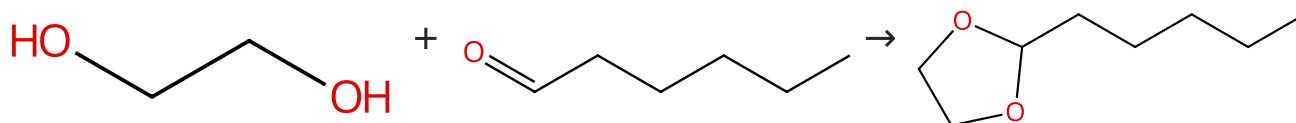
By: Wang, Di; et al

Experimental Protocols

Tetrahedron (2021), 92, 132250.

Scheme 33 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (166)

Suppliers (68)

Suppliers (12)

31-301-CAS-5729818

Steps: 1 Yield: 100%

Distannoxane-catalyzed acetalization of carbonyls.

1.1 Catalysts: Octabutyl-di-μ-hydroxydi-μ₃-oxobis(thiocyanato-κM) tetratin
Solvents: Benzene

By: Otera, Junzo; et al

Tetrahedron (1992), 48(8), 1449-56.

31-301-CAS-4941340

Steps: 1 Yield: 100%

Unusual rate acceleration in the presence of aldehydes of distannoxane-catalyzed acetalization of ketones and deactivated aldehydes: a new mode of carbonyl activation

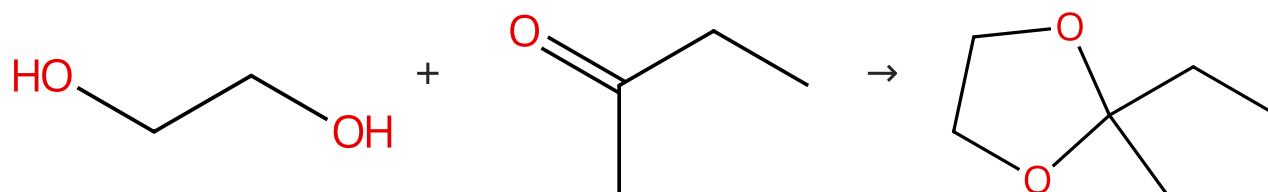
1.1 Catalysts: Octabutyl-di-μ-hydroxydi-μ₃-oxobis(thiocyanato-κM) tetratin
Solvents: Benzene

By: Otera, Junzo; et al

Organometallics (1989), 8(8), 2063-5.

Scheme 34 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (166)

Suppliers (116)

Suppliers (58)

31-301-CAS-1527139

Steps: 1 Yield: 100%

Highly efficient and versatile acetalization of glycol catalyzed by cupric p-toluenesulfonate

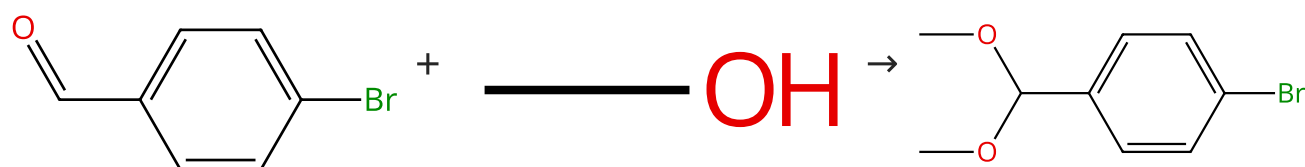
1.1 Catalysts: Benzenesulfonic acid, 4-methyl-, copper(2+) salt (2:1); 90 min, reflux

By: Liu, Cai-Hua; et al

Chinese Journal of Chemistry (2007), 25(1), 83-85.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (91)

Suppliers (444)

Suppliers (61)

31-301-CAS-16699034

Steps: 1 Yield: 100%

Zeolite nanofiber assemblies as acid catalysts with high activity for the acetalization of carbonyl compounds with alcohols

1.1 12 h, 50 °C

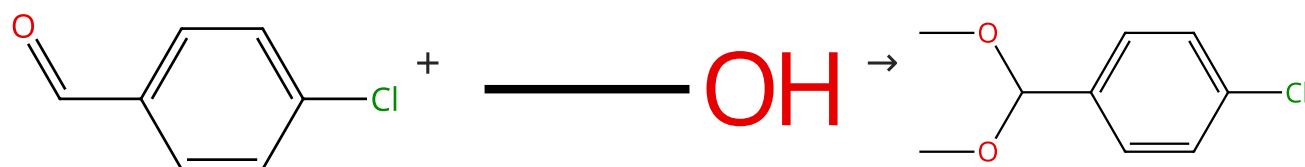
Experimental Protocols

By: Liu, Taotao; et al

RSC Advances (2014), 4(35), 18217-18221.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (99)

Suppliers (444)

Suppliers (44)

31-301-CAS-16699257

Steps: 1 Yield: 100%

Zeolite nanofiber assemblies as acid catalysts with high activity for the acetalization of carbonyl compounds with alcohols

1.1 12 h, 50 °C

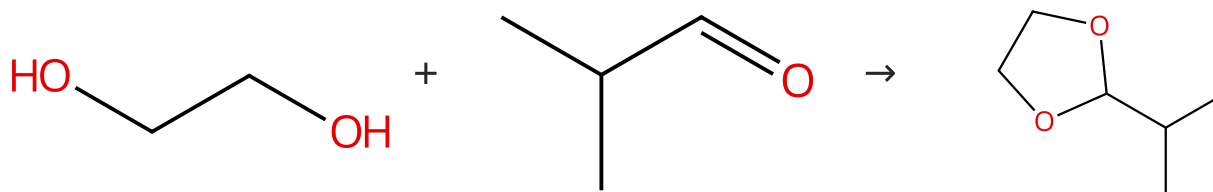
Experimental Protocols

By: Liu, Taotao; et al

RSC Advances (2014), 4(35), 18217-18221.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (166)

Suppliers (74)

Suppliers (12)

31-301-CAS-14902868

Steps: 1 Yield: 100%

Highly efficient and versatile acetalization of glycol catalyzed by cupric p-toluenesulfonate

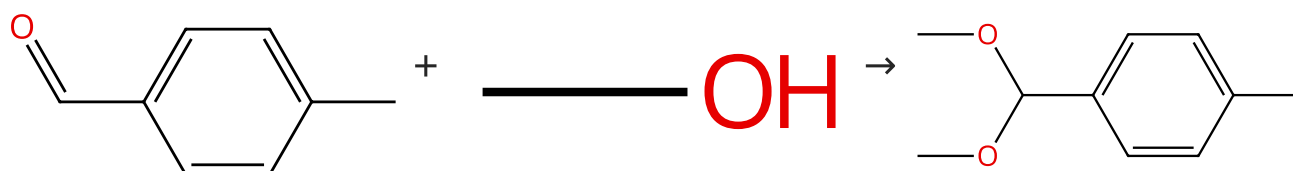
1.1 Catalysts: Benzenesulfonic acid, 4-methyl-, copper(2+) salt (2:1); 90 min, reflux

By: Liu, Cai-Hua; et al

Chinese Journal of Chemistry (2007), 25(1), 83-85.

Scheme 38 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (104)

Suppliers (444)

Suppliers (21)

31-301-CAS-16699259

Steps: 1 Yield: 100%

Zeolite nanofiber assemblies as acid catalysts with high activity for the acetalization of carbonyl compounds with alcohols

1.1 12 h, 50 °C

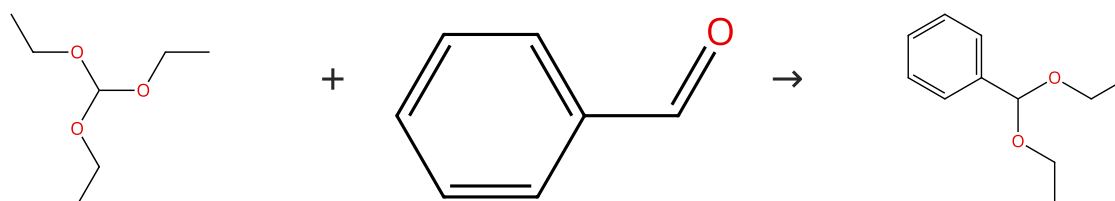
Experimental Protocols

By: Liu, Taotao; et al

RSC Advances (2014), 4(35), 18217-18221.

Scheme 39 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (71)

Suppliers (69)

Suppliers (62)

31-614-CAS-43501688

Steps: 1 Yield: 100%

Mg(OTf)₂.SiO₂: a heterogeneous, recyclable catalyst for the synthesis of acetals from aldehydes or ketones

 1.1 Reagents: *p*-Toluenesulfonic acid
 Catalysts: Magnesium triflate (supported on silica)
 Solvents: Ethanol; 30 min, 30 °C

By: More, Mahesh; et al

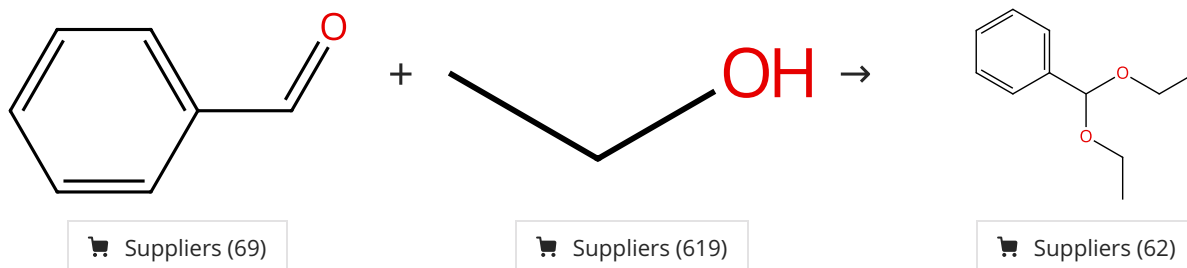
Heterocyclic Letters (2024), 14(4), 787-795.

Experimental Protocols

31-301-CAS-10432178	Steps: 1 Yield: 100%	Green Synthesis of Acetals/Ketals: Efficient Solvent-Free Process for the Carbonyl/Hydroxyl Group Protection Catalyzed by SBA-15 Materials
1.1 Solvents: Ethanol; 30 min, 353 K		By: Perez-Mayoral, Elena; et al
Experimental Protocols		Topics in Catalysis (2009), 52(1-2), 148-152.

Scheme 40 (1 Reaction)

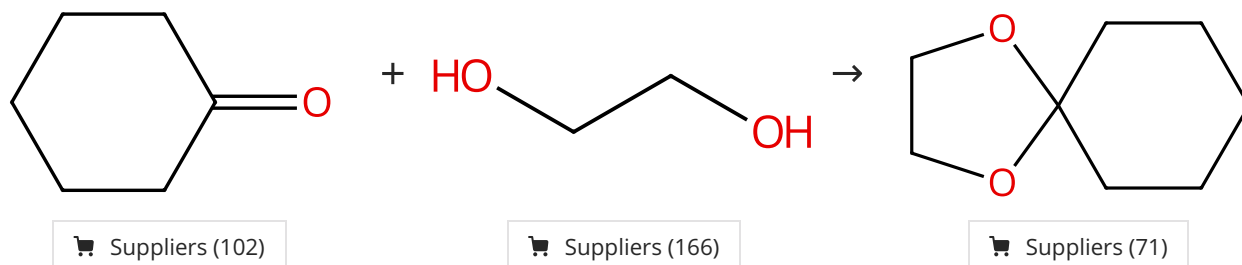
Steps: 1 Yield: 100%



31-301-CAS-9898515	Steps: 1 Yield: 100%	2,4,4,6-Tetrabromo-2,5-cyclohexadienone (TABCO) as a versatile, efficient, and chemoselective catalyst for the acetalization and transacetalization of carbonyl compounds, the preparation of acetonides from epoxides and acylals (1, 1-diacetates) from aldehydes
1.1 Reagents: Triethyl orthoformate Catalysts: 2,4,4,6-Tetrabromo-2,5-cyclohexadien-1-one Solvents: Ethanol; 60 min, 40 °C		By: Firouzabadi, Habib; et al
		Bulletin of the Chemical Society of Japan (2002), 75(10), 2195-2205.

Scheme 41 (2 Reactions)

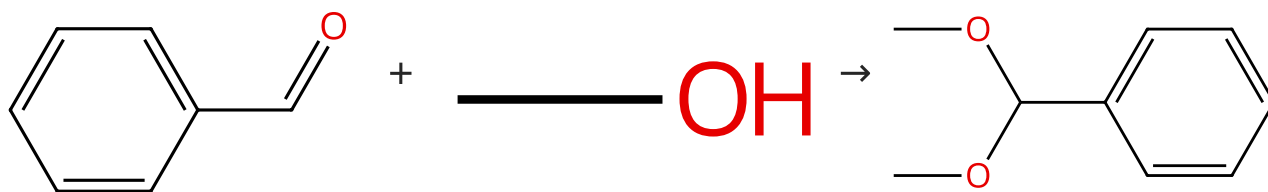
Steps: 1 Yield: 100%



31-301-CAS-15181696	Steps: 1 Yield: 100%	Highly efficient and versatile acetalization of glycol catalyzed by cupric p-toluenesulfonate
1.1 Catalysts: Benzenesulfonic acid, 4-methyl-, copper(2+) salt (2:1) Solvents: Cyclohexane; 90 min, reflux		By: Liu, Cai-Hua; et al
		Chinese Journal of Chemistry (2007), 25(1), 83-85.
31-301-CAS-9209704	Steps: 1 Yield: 100%	1,3-Dioxolanes from carbonyl compounds over zeolite HSZ-360 as a reusable, heterogeneous catalyst
1.1 Solvents: Toluene		By: Ballini, Roberto; et al
		Tetrahedron Letters (1998), 39(12), 1615-1618.

Scheme 42 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (69)

Suppliers (444)

Suppliers (89)

31-301-CAS-16699258

Steps: 1 Yield: 100%

1.1 12 h, 50 °C

Experimental Protocols

Zeolite nanofiber assemblies as acid catalysts with high activity for the acetalization of carbonyl compounds with alcohols

By: Liu, Taotao; et al

RSC Advances (2014), 4(35), 18217-18221.

31-301-CAS-5259979

Steps: 1 Yield: 100%

1.1 **Reagents:** Trimethyl orthoformate
Catalysts: Lithium tetrafluoroborate
Solvents: Methanol; 30 min, reflux

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water

Experimental Protocols

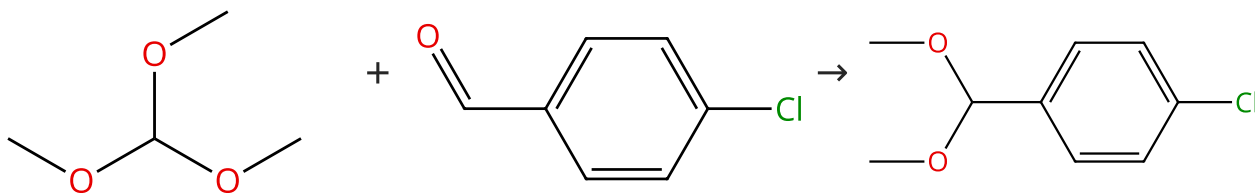
An efficient and versatile procedure for the synthesis of acetals from aldehydes and ketones catalyzed by lithium tetrafluoroborate

By: Hamada, Nao; et al

Synlett (2004), (6), 1074-1076.

Scheme 43 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (68)

Suppliers (99)

Suppliers (44)

31-301-CAS-23963524

Steps: 1 Yield: 100%

1.1 **Catalysts:** Poly(vinylpyrrolidone) hydrogen triiodide
Solvents: Methanol; 3 h, 30 °C

Experimental Protocols

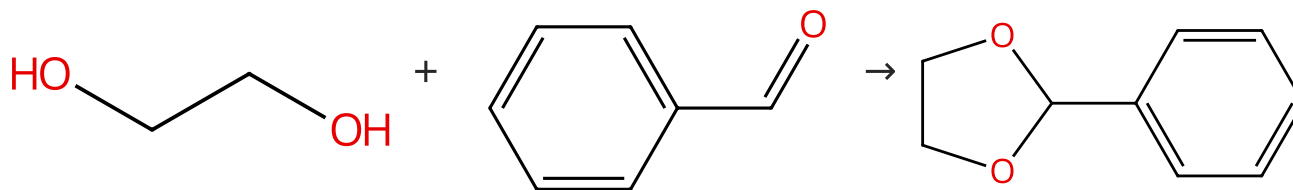
Practical acetalization and transacetalization of carbonyl compounds catalyzed by recyclable PVP-I

By: Wang, Di; et al

Tetrahedron (2021), 92, 132250.

Scheme 44 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (166)

Suppliers (69)

Suppliers (53)

31-301-CAS-2705176	Steps: 1 Yield: 100%	Facile synthesis of MIL-100(Fe) under HF-free conditions and its application in the acetalization of aldehydes with diols
1.1 Catalysts: Amberlyst 15 Solvents: Cyclohexane; 120 min, 80 °C	By: Zhang, Fumin; et al	Chemical Engineering Journal (Amsterdam, Netherlands) (2015), 259, 183-190.
Experimental Protocols		
31-301-CAS-8878007	Steps: 1 Yield: 100%	Unusual rate acceleration in the presence of aldehydes of distannoxane-catalyzed acetalization of ketones and deactivated aldehydes: a new mode of carbonyl activation
1.1 Catalysts: Octabutyl-di- μ -hydroxy-di- μ_3 -oxobis(thiocyanato- κ M) tetratin Solvents: Benzene	By: Otera, Junzo; et al	Organometallics (1989), 8(8), 2063-5.